REGULARITY OF OPTIMISATION OBJECTIVES IN VARIATIONAL INFERENCE

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Abstract. Approximating probability distributions is an important numerical procedure with applications in computational statistics and machine learning. This problem can be converted to an optimisation problem of minimising a statistical divergence on the parameter space of a family of distributions, known as variational inference. In this report we survey existing results and propose new ones, on the regularity of variational optimisation objectives, necessary for downstream guarantees on the convergence of variational methods.

1. Preliminaries

Variational inference converts the problem of approximating a probability distribution into an optimisation problem. In black box variational inference, we choose a family of distributions $\mathcal{Q}$ and try to find the distribution $Q^* \in \mathcal{Q}$ that minimises a divergence (often KL) from the target distribution π. Variants of stochastic gradient descent are generally used to solve the resulting optimisation problem, which is non-convex in general.

Definition 1.1 (Location-scale family)**.** *A family of distributions is location-scale if we can write every member as the result of an affine transformation of a fixed base distribution.*

$$z \sim \mathit{LocScale}(m, C, q) \iff z \overset{d}{=} m + Cu, u \sim q$$

where $m \in \mathbb{R}^d$ is the location parameter, $C \in \mathbb{R}^{d\times d}$ is the scale parameter and q is the base distribution. Letting $\lambda = (m, C)$, we can write q_λ to denote $\mathit{LocScale}(m, C, q)$, with the density

$$q_\lambda(z) = \frac{1}{|\det C|} q\left(C^{-1}(z - m)\right).$$

A special case is the mean-field family, which gives us a diagonal Cholesky factor from any real valued vector. $C = D_\phi(s)$, where the function $\phi : \mathbb{R} \to \mathbb{R}_+$ is called the diagonal conditioner, $D_\phi(s) := \mathit{diag}(\phi(s_1), \ldots, \phi(s_d))$ and we define $\lambda = (m, s)$.

The simplest example is multivariate Gaussian distributions, which are location-scale with the standard normal $q = \mathcal{N}(0, I)$ as the base distribution, and $q_\lambda \sim \mathcal{N}(m, CC^\top)$.

The standard variational objective (known as negative ELBO) is equivalent to minimising the KL divergence and is derived below. Let π be the target density (upto an unknown multiplicative constant), Λ be the parameter space, and q_λ for $\lambda \in \Lambda$ be a member of the variational family $\mathcal{Q}$.

$$\begin{aligned} D_{KL}(q_\lambda \| \pi) &= \int q_\lambda(x) \log \frac{q_\lambda(x)}{\pi(x)} dx \\ &= \int q_\lambda(x) \log q_\lambda(x) dx - \int q_\lambda(x) \log \pi(x) dx \\ &= \mathbb{E}_{x\sim q_\lambda}[-\log \pi(x)] + \mathbb{E}_{x\sim q_\lambda}[\log q_\lambda(x)] \\ &:= l(\lambda) - h(\lambda) \end{aligned}$$

The first term $l(\lambda)$ is called the energy, while the second term $h(\lambda)$ is the entropy of q_λ.

2. Regularity of the entropy term

Definition 2.1 (Entropy)**.** *The entropy of a probability distribution with density q is defined as*

$$H[q] = -\int q(x) \log q(x)\, dx.$$

2.1. **Concavity.**

Lemma 2.2 (Concavity of entropy)**.** *The entropy functional H is concave: for any two densities p, q and $t \in [0,1]$,*

$$H[tp + (1-t)q] \geq tH[p] + (1-t)H[q].$$

Proof. The function $\phi : x \mapsto -x \log x$ is concave on $[0, \infty)$ since $\phi''(x) = -1/x < 0$ for $x > 0$. For the mixture density $r = tp + (1-t)q$, applying concavity of ϕ pointwise gives

$$\begin{aligned} H[r] = \int \phi(r(x))\, dx &= \int \phi(tp(x) + (1-t)q(x))\, dx \\ &\geq \int \big[t\phi(p(x)) + (1-t)\phi(q(x))\big]\, dx \\ &= tH[p] + (1-t)H[q]. \end{aligned}$$

□

Lemma 2.3. *The entropy of a location-scale distribution q_λ can be expressed in terms of the entropy of the base distribution q as*

$$h(\lambda) = H[q_\lambda] = H[q] + \log|\det C|.$$

with gradient $\nabla h(\lambda) = (0, C^{-\top})$.

Proof. By Definition 1.1, $z = m + Cu$ with $u \sim q$, so q_λ has density $q_\lambda(z) = \frac{1}{|\det C|} q(C^{-1}(z-m))$. The differential entropy of q_λ is

$$\begin{aligned} H[q_\lambda] &= -\int q_\lambda(z) \log q_\lambda(z)\, dz \\ &= -\int q_\lambda(z) \log\left(\frac{1}{|\det C|} q(C^{-1}(z-m))\right) dz \\ &= \log|\det C| - \int q_\lambda(z) \log q(C^{-1}(z-m))\, dz. \end{aligned}$$

Substituting $u = C^{-1}(z-m)$, so $dz = |\det C|\, du$ and $q_\lambda(Cu+m) = \frac{1}{|\det C|} q(u)$, the remaining integral becomes

$$\int \frac{1}{|\det C|} q(u) \log q(u) \cdot |\det C|\, du = \int q(u) \log q(u)\, du = -H[q].$$

Therefore $h(\lambda) = H[q_\lambda] = H[q] + \log|\det C|$.

Since $H[q]$ is constant in $\lambda = (m, C)$, the gradient with respect to m is zero. For the gradient with respect to C, using $\nabla_C \log|\det C| = C^{-\top}$, we obtain $\nabla h(\lambda) = (0, C^{-\top})$. □

Thus, for the entropy $h(\lambda)$ of a location-scale family to be concave, we need $C \mapsto \log|\det C|$ to be concave. Thus we need to restrict C to the set of strictly positive definite matrices, which is satisfied by the mean-field parametrisation we saw above.

2.2. **Smoothness.**

Theorem 2.4 (Lemma 1 of [KOWMG24])**.** *If the log diagonal conditioner $\log \phi$ is L_h-smooth, then the entropy term $h(\lambda)$ is L_h-smooth.*

Proof.

$$\begin{aligned} h(\lambda) = H[q_\lambda] &= H[q] + \log|\det C| \\ &= H[q] + \sum_{i=1}^{d} \log \phi(s_i) \\ \nabla^2 h(\lambda) &= \operatorname{diag}\left(\frac{\partial^2 h}{\partial s_1}, \ldots, \frac{\partial h^2}{\partial s_d}\right) \\ &= \operatorname{diag}\left(\frac{\partial^2 \log\phi}{\partial s_1^2}, \ldots, \frac{\partial^2 \log\phi}{\partial s_d^2}\right) \preceq L_h I. \end{aligned}$$

The diagonal entries satisfy $\left|\frac{\partial^2 \log\phi}{\partial s_i^2}\right| \leq L_h$, hence $\|\nabla^2 h\|_2 \leq L_h$. □

3. Regularity of the energy term

Assumption A. The base distribution of the location-scale family, q is *standardised*, i.e. $\mathbb{E}_{u\sim q}[u] = 0$ and $\mathrm{Var}_{u\sim q}[u] = I$.

3.1. Convexity and strong convexity.

Theorem 3.1 (Theorem 9 of [Dom20])**.** *If $-\log\pi$ is convex, then the energy term $l(\lambda)$ is convex. If $-\log\pi$ is α-strongly convex and assumption A holds for the location-scale family $\mathcal{Q}$, then $l(\lambda)$ is α-strongly convex.*

Proof. For convexity, let $\lambda, \lambda' \in \Lambda$ and $t \in [0,1]$. Define $T_\lambda(u) = Cu + m$, i.e. for fixed u, $T_\lambda(u)$ is an affine function of λ. By convexity of $f = -\log\pi$, for each u we have:

$$\begin{aligned} f(T_{t\lambda+(1-t)\lambda'}(u)) &\leq tf(T_\lambda(u)) + (1-t)f(T_{\lambda'}(u)) \\ \mathbb{E}_{u\sim q}[f(T_{t\lambda+(1-t)\lambda'}(u))] &\leq \mathbb{E}_{u\sim q}[tf(T_\lambda(u)) + (1-t)f(T_{\lambda'}(u))] \\ l(t\lambda + (1-t)\lambda') &\leq tl(\lambda) + (1-t)l(\lambda'). \end{aligned}$$

For strong convexity, we can write $f(x) = f_0(x) + \frac{\alpha}{2}\|x\|_2^2$ for some convex f_0. Then, $l(\lambda) = l_0(\lambda) + \frac{\alpha}{2}\mathbb{E}_{x\sim q}\|x\|_2^2$ where $l_0(x) = \mathbb{E}_{x\sim q_\lambda}[f_0(x)]$ is convex by the previous argument.

$$\begin{aligned} \mathbb{E}_{x\sim q_\lambda}\|x\|_2^2 &= \mathbb{E}_{u\sim q}\|Cu + m\|_2^2 \\ &= \sum_{i=1}^{d} \mathbb{E}[s_i^2 u_i^2 + 2s_i u_i m_i + m_i^2] \\ &= \sum_{i=1}^{d} s_i^2 \mathbb{E}[u_i^2] + 2s_i m_i \mathbb{E}[u_i] + m_i^2 \\ &= \sum_{i=1}^{d} s_i^2 + m_i^2 \\ &= \|\lambda\|_2^2 \end{aligned}$$

For more general parametrisations we can show $\mathbb{E}_{u\sim q}\|Cu + m\|_2^2 = \|C\|_F^2 + \|m\|^2 = \|\lambda\|^2$.

Hence we have that $l(\lambda) = l_0(\lambda) + \frac{\alpha}{2}\|\lambda\|_2^2$ is α-strongly convex. □

3.2. Smoothness.

Theorem 3.2 (Theorem 1 of [Dom20])**.** *Let $-\log\pi$ be L_ℓ-smooth, and let the variational family be the location-scale family of Definition 1.1 with a standardised base distribution q and a linear conditioner $\phi(s) = s$. Then the energy $l(\lambda)$ is L_ℓ-smooth.*

4. Convexity of the variational objective for general f-divergences

This section of the survey consists of original work. We attempt to find conditions to characterise the convexity of the variational objective for general f divergences (rather than just KL divergence), first proposed as a VI method in [WLH21].

Definition 4.1 (f-divergence)**.** *Given a convex function (called generator) $f : (0,\infty) \to \mathbb{R}$ such that $f(1) = 0$, the f-divergence between two probability distributions P and Q with densities p and q is defined as*

$$D_f(P\|Q) = \int q(x) f\left(\frac{p(x)}{q(x)}\right) dx.$$

This satisfies $D_f(P\|Q) \geq 0$ with equality if and only if $P = Q$. It is not in general a distance metric since it is not symmetric and does not satisfy the triangle inequality.

The most common example of a f-divergence is the aforementioned Kullback-Leibler (KL) divergence, which corresponds to $f(t) = t\log t$.

Theorem 4.2 (Convexity of reverse f-divergence for location-scale families)**.** *Let $f : (0,\infty) \to \mathbb{R}$ be convex with $f(1) = 0$, and define*

$$\widehat{f}(s) \;:=\; f(e^s)\, e^{-s}, \qquad s \in \mathbb{R}.$$

Let π be a target density on $\mathbb{R}^d$ with $-\log \pi$ convex. Let the variational family be location scale:

$$\mathcal{Q} \;=\; \big\{ q_\lambda(x) = |\det C|^{-1}\, q_0\big(C^{-1}(x-m)\big) : \lambda = (m, C) \in \mathbb{R}^d \times \mathcal{C} \big\}$$

with fixed base density q_0, where $\mathcal{C}$ is a convex set of invertible matrices on which $C \mapsto -\log|\det C|$ is convex (e.g. the PD cone, or lower-triangular matrices with positive diagonal as in mean-field parametrisation).

If $\widehat{f}$ is convex and non-decreasing on $\mathbb{R}$, then $\lambda \longmapsto D_f(q_\lambda \,\|\, \pi)$ is convex on $\mathbb{R}^d \times \mathcal{C}$.

Proof. Substituting $x = Cu + m$, $dx = |\det C|\, du$, and using $q_\lambda(Cu+m) = q_0(u)/|\det C|$:

$$\begin{aligned}
D_f(q_\lambda \,\|\, \pi) &= \int f\left(\frac{q_\lambda(x)}{\pi(x)}\right) \pi(x)\, dx \\
&= \int f\left(\frac{q_0(u)}{|\det C| \pi(Cu+m)}\right) \pi(Cu+m)\, |\det C|\, du \\
&= \int f\left(\frac{q_0(u)}{|\det C| \pi(Cu+m)}\right) \frac{\pi(Cu+m)\, |\det C|}{q_0(u)} q_0(u)\, du \\
&:= \int f\left(r(\lambda)\right) \frac{1}{r(\lambda)} q_0(u)\, du \\
&= \int \widehat{f}(\log r(\lambda))\, q_0(u)\, du, \\
&= \mathbb{E}_{u \sim q_0}\big[\widehat{f}(\log r_\lambda(u))\big],
\end{aligned}$$

We have

$$\log r_\lambda(u) \;=\; \log q_0(u) \;-\; \log|\det C| \;-\; \log \pi(Cu+m).$$

Term by term in $\lambda = (m, C)$:

(1) $\log q_0(u)$ is independent of λ.
(2) $-\log|\det C|$ is convex on $\mathcal{C}$ by hypothesis.
(3) $\lambda \mapsto Cu + m$ is affine for each fixed u; since $-\log\pi$ is convex, the composition $\lambda \mapsto -\log\pi(Cu+m)$ is convex.

Hence $\lambda \mapsto \log r_\lambda(u)$ is convex on $\mathbb{R}^d \times \mathcal{C}$ for each u. By hypothesis $\widehat{f}$ is convex and non-decreasing. Composition of a convex non-decreasing function with a convex function is convex, so $\lambda \longmapsto \widehat{f}(\log r_\lambda(u))$ is convex on $\mathbb{R}^d \times \mathcal{C}$ for each u. Since q_0 does not depend on λ, taking expectation preserves convexity, giving the result. □

4.1. **Examples.**

- **Reverse KL** ($f(t) = t\log t$): $\widehat{f}(s) = s$, linear hence convex non-decreasing. Recovers Domke's Theorem 9.
- **χ^2-divergence**: the canonical generator $f(t) = (t-1)^2$ gives $\widehat{f}(s) = 2\cosh(s) - 2$, which is convex but *not* monotone, so the theorem does not apply in this form. See next sections.

4.2. **Equivalence class of generators and existence of a convexity-enabling shift.**

Lemma 4.3 (Equivalence class of generators)**.** *f-divergences can be defined using multiple generators- they are invariant under $f(t) \mapsto f_c(t) = f(t) + c(t-1)$ for any $c \in \mathbb{R}$, i.e.*

$$D_{f_c}(q \,\|\, \pi) = D_f(q \,\|\, \pi)$$

Proof.

$$\begin{aligned} D_{f_c}(q \,\|\, \pi) &= \int f_c(q/\pi)\, \pi \, dx \\ &= \int \big(f(q/\pi) + c(q/\pi - 1)\big)\pi \, dx \\ &= D_f(q \,\|\, \pi) + c \int (q/\pi - 1)\, \pi \, dx \\ &= D_f(q \,\|\, \pi) + c \int q(x)dx - c \int \pi(x)dx \\ &= D_f(q \,\|\, \pi). \end{aligned}$$

□

The shift $c(t-1)$ does not change the divergence value. However, the hypothesis on $\widehat{f}$ in theorem 4.2 is sensitive to this choice,

$$\widehat{f}_c(s) = \widehat{f}(s) + c\big(1 - e^{-s}\big).$$

The convexity and monotonicity of $\widehat{f}_c$ depend on the choice of c, which affects the applicability of the theorem. Hence we need to search the equivalence class $\{\varphi + c(t-1)\}_{c\in\mathbb{R}}$ for a representative which satisfies the hypothesis. Using this, we derive a sufficient condition that is independent of the shift.

Lemma 4.4 (Existence of a convexity-enabling shift)**.** *Let $f : (0,\infty) \to \mathbb{R}$ be convex, twice continuously differentiable, with $f(1) = 0$. Define*

$$P_f(t) := tf'(t) - f(t), \qquad Q_f(t) := t^2 f''(t) - tf'(t) + f(t).$$

Then there exists $c \in \mathbb{R}$ such that the equivalent generator $f_c(t) := f(t) + c(t-1)$ has $\widehat{f}_c$ convex and non-decreasing on $\mathbb{R}$ if and only if

$$-\inf_{t>0} P_f(t) \;\le\; \inf_{t>0} Q_f(t).$$

The set of valid shifts is the (possibly degenerate) interval

$$c \;\in\; \big[-\inf_{t>0} P_f(t),\; \inf_{t>0} Q_f(t)\big].$$

Proof. We want for the shifted generator, $\widehat{f}_c$ to be convex and non-decreasing so we compute its derivatives in terms of the original generator:

$$\begin{aligned} \widehat{f}_c(s) &= f_c(e^s)e^{-s} \\ &= \big(f(e^s) + c(e^s - 1)\big)e^{-s} \\ &= (f(e^s) - c)e^{-s} + c \end{aligned}$$

$$\begin{aligned} \widehat{f}_c{}'(s) &= \frac{d}{ds}\big(\widehat{f}_c(s)\big) \\ &= e^{-s}(e^s f'(s)) - e^{-s}(f(e^s) - c) \\ &= e^{-s}(e^s f'(e^s) - f(e^s) + c) \\ &:= \frac{1}{t}(tf'(t) - f(t) + c) \end{aligned}$$

For non-decreasing, we want $\widehat{f}_c{}'(s) \ge 0$ for all s, which is equivalent to $c \ge -P_f(t)$ for all $t > 0$ i.e. $c \ge \sup_{t>0} -P_f(t) = -\inf_{t>0} P_f(t)$.

$$\begin{aligned}
\widehat{f_c}''(s) &= \frac{d}{ds}\big(e^{-s}(e^s f'(e^s) - f(e^s) + c)\big) \\
&= -e^{-s}(e^s f'(e^s) - f(e^s) + c) + e^{-s}\frac{d}{ds}(e^s f'(e^s) - f(e^s) + c) \\
&= -e^{-s}(e^s f'(e^s) - f(e^s) + c) + e^{-s}(e^{2s} f''(e^s) + e^s f'(e^s) - e^s f'(e^s)) \\
&= e^{-s}\big(e^{2s} f''(e^s) - e^s f'(e^s) + f(e^s) - c\big) \\
&:= \frac{1}{t}(t^2 f''(t) - t f'(t) + f(t) - c)
\end{aligned}$$

For convexity, we want $\widehat{f_c}''(s) \geq 0$ for all s, which is equivalent to $c \leq Q_f(t)$ for all $t > 0$, i.e. $c \leq \inf_{t>0} Q_f(t)$. Both conditions hold simultaneously for some c iff the interval is non-empty. □

Lemma 4.5 (Forward f divergence as reverse $f^\star$ divergence)**.** *Let $f : (0, \infty) \to \mathbb{R}$ be convex with $f(1) = 0$. Define the conjugate generator $f^\star(t) := tf(1/t)$. Then*

$$D_f(\pi \| q_\lambda) = D_{f^\star}(q_\lambda \| \pi).$$

Proof.

$$\begin{aligned}
D_f(\pi \| q_\lambda) &= \int q_\lambda(x) f\left(\frac{\pi(x)}{q_\lambda(x)}\right) dx \\
&= \int \pi(x) \cdot \frac{q_\lambda(x)}{\pi(x)} f\left(\frac{\pi(x)}{q_\lambda(x)}\right) dx \\
&= \int \pi(x) \cdot f^*\left(\frac{q_\lambda(x)}{\pi(x)}\right) dx \\
&= D_{f^\star}(q_\lambda \| \pi).
\end{aligned}$$

□

Corollary 4.6 (Convexity of forward f-divergence for location-scale families)**.** *Under the setup of Theorem 4.2, suppose $f : (0, \infty) \to \mathbb{R}$ is convex with $f(1) = 0$. Then*

$$\lambda \longmapsto D_f(\pi \,\|\, q_\lambda)$$

is convex on $\mathbb{R}^d \times \mathcal{C}$ provided

1. *$f \circ \exp$ is convex on $\mathbb{R}$ (i.e. f is log-convex in its argument), and*
2. *f is non-increasing on $(0, \infty)$.*

Equivalently, the conjugate generator $f^\star(t) := tf(1/t)$ satisfies the hypothesis of Theorem 4.2.

Proof. By lemma 4.5 $D_f(\pi \| q_\lambda) = D_{f^\star}(q_\lambda \| \pi)$ with $f^\star(t) = tf(1/t)$, we can apply Theorem 4.2 to $f^\star$.

$$\begin{aligned}
\widehat{f^\star}(s) &= f^*(e^s)e^{-s} \\
&= e^s f(1/e^s)e^{-s} \\
&= f(e^{-s}).
\end{aligned}$$

Hence the required hypothesis "$\widehat{f^\star}$ convex and non-decreasing" translates to (i) and (ii) above. □

Proposition 4.7 (Mutual exclusivity of forward/reverse hypotheses)**.** *Let $f : (0, \infty) \to \mathbb{R}$ be a non-trivial convex function with $f(1) = 0$. Then, the convexity hypothesis of Theorem 4.2 can be satisfied for at most one of $D_f(q_\lambda \| \pi)$ and $D_f(\pi \| q_\lambda)$.*

Proof. We translate both hypotheses into pointwise conditions on f. Setting $t = e^s$, we get (see proof of lemma 4.4)

$$\begin{aligned}
\widehat{f}'(s) &= \frac{1}{t}\big[tf'(t) - f(t)\big] = \frac{P_f(t)}{t}, \\
\widehat{f}''(s) &= \frac{1}{t}\big[t^2 f''(t) - tf'(t) + f(t)\big] = \frac{Q_f(t)}{t}.
\end{aligned}$$

Hence the reverse hypothesis (Theorem 4.2) is equivalent to

$$P_f(t) \geq 0 \quad \text{and} \quad Q_f(t) \geq 0 \qquad \forall\, t > 0. \tag{R}$$

By Corollary 4.6, the forward hypothesis is equivalent to

$$f \text{ non-increasing} \quad \text{and} \quad f \circ \exp \text{ convex}. \tag{F}$$

Assume both (R) and (F) hold. From f non-increasing and $f(1) = 0$,

$$f \geq 0 \text{ on } (0, 1], \qquad f \leq 0 \text{ on } [1, \infty). \tag{$*$}$$

Now, note that

$$P_f'(t) = f'(t) + tf''(t) - f'(t) = tf''(t) > 0$$

and thus $P_f(t)$ is monotone non-decreasing in t. Hence

$$\inf_{t>0} P_f(t) = \lim_{t \to 0^+} P_f(t) \geq 0 \text{ by (R)}.$$

We now evaluate this limit. By convexity, for any $s \in (0, t)$,

$$\begin{aligned} f(s) &\geq f(t) + f'(t)(s - t) \\ f'(t)(s - t) &\leq f(s) - f(t) \\ -tf'(t) &\leq f(0^+) - f(t) \qquad (s \to 0^+). \end{aligned}$$

Since f is non-increasing by (F), $f'(t) \leq 0$, hence $tf'(t) \leq 0$. Combining,

$$f(t) - f(0^+) \;\leq\; tf'(t) \;\leq\; 0.$$

Letting $t \to 0^+$, both bounds vanish, so $tf'(t) \to 0$ and thus

$$\lim_{t \to 0^+} P_f(t) \;=\; \lim_{t \to 0^+} \big[tf'(t) - f(t)\big] \;=\; -f(0^+).$$

Now by the above limit and (R),

$$-f(0^+) \;\geq\; 0 \quad \Longrightarrow \quad f(0^+) \leq 0$$

and combined with $(*)$ we get $f(0^+) = 0$.

Now f is convex with $f(0^+) = f(1) = 0$ and $f \geq 0$ on $(0, 1]$, so

$$f \equiv 0 \quad \text{on } (0, 1]$$

Now since f is constant to the left if 1, $f'_-(1) = 0$, and convexity at $t = 1$ gives $f'_+(1) \geq f'_-(1) = 0$. But f non-increasing in $\implies f'_+(1) \leq 0$. So $f'_+(1) = 0$.

Then, by convexity on $t \in [1, \infty)$, we have

$$f(t) \geq f(1) + f'_+(1)(t - 1) = 0 + 0 \cdot (t - 1) = 0$$

But $(*)$ also gives $f \leq 0$ on $[1, \infty)$, so

$$f \equiv 0 \quad \text{on } [1, \infty)$$

Together $f \equiv 0$ on $(0, \infty)$, contradicting non-triviality. □

4.3. **Application to standard divergences.**

4.3.1. *Reverse KL divergence.*

$$\begin{aligned} f(t) &= t\log t \\ P_f(t) &= tf'(t) - f(t) = t\cdot(1+\log t) - t\log t = t \\ \inf_{t>0} P_f(t) &= 0 \\ Q_f(t) &= t^2 f''(t) - tf'(t) + f(t) = t^2\cdot\frac{1}{t} - t\cdot(1+\log t) + t\log t = 0 \end{aligned}$$

Thus the unique valid shift is $c = 0$.

4.3.2. χ^2 *divergence.*

$$\begin{aligned} f(t) &= (t-1)^2 \\ P_f(t) &= tf'(t) - f(t) = 2t(t-1) - (t-1)^2 = t^2 - 1 \\ \inf_{t>0} P_f(t) &= -1 \\ -\inf_{t>0} P_f(t) &= 1 \\ Q_f(t) &= t^2 f''(t) - tf'(t) + f(t) = 2t^2 - 2t(t-1) + (t-1)^2 = t^2 + 1 \\ \inf_{t>0} Q_f(t) &= 1 \end{aligned}$$

Thus the unique valid shift is $c = 1$.

4.3.3. α*-divergences.* The canonical generator for the α-divergence (for $\alpha \in \mathbb{R}\setminus\{0,1\}$) is

$$f(t) = \frac{t^\alpha - 1}{\alpha(\alpha-1)}, \qquad f'(t) = \frac{t^{\alpha-1}}{\alpha-1}, \qquad f''(t) = t^{\alpha-2}.$$

$$\begin{aligned} P_f(t) &= tf'(t) - f(t) = \frac{t^\alpha}{\alpha-1} - \frac{t^\alpha - 1}{\alpha(\alpha-1)} = \frac{t^\alpha}{\alpha} + \frac{1}{\alpha(\alpha-1)}, \\ Q_f(t) &= t^2 f''(t) - tf'(t) + f(t) = t^\alpha - \frac{t^\alpha}{\alpha-1} + \frac{t^\alpha - 1}{\alpha(\alpha-1)} = \frac{\alpha-1}{\alpha}\,t^\alpha - \frac{1}{\alpha(\alpha-1)}. \end{aligned}$$

For $\alpha > 1$, both $t \mapsto t^\alpha/\alpha$ and $t \mapsto \frac{\alpha-1}{\alpha}t^\alpha$ are non-negative and vanish as $t \to 0^+$, so

$$\inf_{t>0} P_f(t) \;=\; \frac{1}{\alpha(\alpha-1)}, \qquad \inf_{t>0} Q_f(t) \;=\; -\frac{1}{\alpha(\alpha-1)}.$$

Thus, the unique valid shift is $c = -\frac{1}{\alpha(\alpha-1)}$.

For example, at $\alpha = 2$ this gives $c = -1/2$ and $f_c(t) = \frac{1}{2}t(t-1)$, which agrees up to a factor of 2 with the shifted χ^2 generator $f_1(t) = t(t-1)$ from the previous subsection (consistent, since the two generators define the same divergence up to scaling).

For $0 < \alpha < 1$, $\alpha(\alpha-1) < 0$, so $\inf P_f = \frac{1}{\alpha(\alpha-1)} < 0$ but $\inf Q_f = -\infty$ since $\frac{\alpha-1}{\alpha} < 0$ and thus $\lim_{t\to\infty} \frac{\alpha-1}{\alpha}t \to -\infty$. So no valid shift exists and the theorem does not apply for $\alpha \in (0,1)$.

For $\alpha < 0$, $t^\alpha \to +\infty$ as $t \to 0^+$ and $1/\alpha < 0$, so $\inf_{t>0} P_f(t) = -\infty$ and no bvalid shift exists, so the theorem also does not apply for negative α.

4.3.4. *Forward KL divergence.* We can invoke proposition 4.7 to immediately conclude that the forward KL divergence $D_{KL}(\pi\|q_\lambda)$ does not satisfy the convexity hypothesis of Theorem 4.2, since the reverse KL divergence does.

This negative result is consistent with the well-known qualitative behaviour of forward KL: it is mass-covering rather than mode-seeking, and its sublevel sets in location-scale parameters are not convex in general.

4.3.5. *Total variation distance.* $f(t) = \frac{1}{2}|t-1|$ is not C^2 at $t = 1$, so Lemma 4.4 does not directly apply.

5. Future directions

This work can be extended in several directions.

- **Extension to more general variational families.** The current results are for location-scale families. It would be valuable to extend the analysis to more general variational families.
- **Necessary conditions for convexity of f divergences.** The current results only give sufficient conditions.
- **Conditions for smoothness and strong convexity for general f-divergences.** We have characterised convexity for general f-divergences, but not smoothness or strong convexity.

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